Article 0.10: Theorem of Algorithmic Spectral Completeness (Revised Edition – 33 Problems)

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Abstract

We present the Theorem of Algorithmic Spectral Completeness, a **meta-theorem** (i.e., a theorem about theorems) that unifies the resolution of NP-complete problems and deep mathematical conjectures under the framework of the Spectral Reduction Lemma (Kithima's Lemma). The theorem asserts that any problem or conjecture that admits a spectral encoding satisfying the four pillars (P1–P4) is decidable by a finite deterministic algorithm – the D-RSP algorithm – which runs in time polynomial in the size of the SAT encoding. The algorithm provides an explicit constructive certificate (solution, counterexample, or proof of impossibility). The proof of this meta-theorem is included here; it follows directly from the Spectral Reduction Lemma and the properties of the D-RSP extractor. The theorem is then instantiated for the **thirty-three resolved problems** (including BSD, Fuglede, Barnette and prime families). We provide a detailed section explaining how each of the thirty-three conjectures is proved via the spectral reduction lemma, and the role of the Algorithmic Spectral Completeness Theorem in these proofs. This article serves as the logical capstone of the foundational series.

Contents

1 Introduction

The Spectral Reduction Lemma (Article0.9) provides an operational tool: given a problem encoded as a spectral Hamiltonian satisfying the four pillars, the D-RSP algorithm decides it. The Theorem of Algorithmic Spectral Completeness elevates this tool to a meta-statement: **any** conjecture that can be captured by such an encoding is automatically decidable by a uniform algorithm. This theorem is not a conjecture about a specific problem but a statement about the entire class of problems that admit a spectral encoding. Its proof is constructive and relies solely on the already established lemmas of the spectral framework.

2 What is a meta-theorem?

A meta-theorem is a theorem that speaks about other theorems or about the properties of a formal system. In our context, the Theorem of Algorithmic Spectral Completeness does not directly prove any particular conjecture; rather, it establishes a **criterion** for decidability. It says:

> *"If a conjecture X can be encoded in the spectral framework (i.e., there exists a SAT instance Φ_X and a Hamiltonian H_X satisfying P1–P4 such that the satisfiability of Φ_X is equivalent to the truth of X), then there exists a finite deterministic algorithm that decides X and produces a certificate."*

This is a statement **about** the spectral encoding, not a statement within the encoding. Its proof is given below and is independent of any particular X.

3 Theorem of Algorithmic Spectral Completeness

Theorem 3.1 (Algorithmic Spectral Completeness (Kithima, 2026))**.** *Let X be a mathematical conjecture (or a decision problem) for which there exists a spectral encoding:*

- *A Hamiltonian $H = \hat{V} + \Delta$ on a hypercube (or a constant-degree expander) and a SAT instance Φ such that:*
  1. *Satisfying assignments of Φ correspond bijectively to solutions/certificates of X (or to the absence of a counterexample, depending on the strategy).*
  2. *The four pillars are verified:*
     - ***P1**: Unique strictly positive ground state ψ_0.*
     - ***P2**: Constant spectral gap $\gamma(H) \geq 2$ (or polynomial but bounded below by a positive constant).*
     - ***P3**: Logarithmic area law, implying an MPS representation of ψ_0 with bond dimension polynomial in the size of Φ.*
     - ***P4**: The deterministic D-RSP algorithm extracts a configuration (or proves unsatisfiability) in a finite number of steps polynomial in the size of Φ.*

Then there exists a finite deterministic algorithm that, according to one of the four core strategies (CD, EB, FV, FD) – or, more generally, any strategy that produces such an encoding – decides the truth of X and, if applicable, produces an explicit certificate (a solution, a counterexample, or a proof of impossibility). The algorithm runs in time polynomial in the size of the SAT instance Φ. If the size of Φ is exponential in the original input size (e.g., for the Hadamard conjecture), the algorithm remains theoretically finite, with a complexity exponential in the input parameter – this is acceptable for a constructive existence proof.

3.1 Proof of the meta-theorem

The proof is a direct consequence of the Spectral Reduction Lemma (Article0.9) and the definition of the D-RSP algorithm (Article0.4). By assumption, the spectral encoding satisfies P1–P4. The D-RSP algorithm, when applied to the Hamiltonian H, terminates after a finite number of steps polynomial in the size of Φ. It returns a satisfying assignment of Φ if and only if such an assignment exists. By the bijection between satisfying assignments and the truth of X (or the existence of a counterexample), the algorithm decides X. The certificate is extracted from the assignment. This argument holds for any such X; hence the theorem is a meta-theorem. The proof does not rely on any specific feature of X beyond the existence of the encoding.

3.2 Remarks on the theorem

1. **Meta-nature**: The theorem is not proved by a single instance but by a generic argument that applies to all problems satisfying the hypotheses. It is a theorem **about** the spectral method, not a theorem within it.

2. **"Theoretical" algorithm**: The algorithm exists in principle; its actual execution may require astronomical constants, but its existence is mathematically guaranteed.

3. **“Finite” algorithm**: The algorithm terminates after a bounded number of steps, where the bound depends on the size of Φ.

4. **Polynomial time relative to the SAT instance**: This is the natural measure because the D-RSP algorithm works on the hypercube of dimension M. For problems where the original input size is much smaller (e.g., a number n given in binary), the encoding may produce an instance of size exponential in the input size (e.g., $M = n^2$ for Hadamard). In such cases, the algorithm runs in time polynomial in n, which is exponential in $\log n$. The theorem does not claim polynomial time in the input size; it claims polynomial time in the SAT instance size.

5. **Constructivity**: The certificate is explicitly extracted from the ground state. No non-constructive existence argument is used.

4 Role of the Algorithmic Spectral Completeness Theorem in proving the thirty-three conjectures

The Algorithmic Spectral Completeness Theorem is not merely a meta-statement; it plays a crucial role in the proof of each of the thirty-three conjectures. Specifically:

- For each conjecture, we construct an explicit spectral encoding (SAT instance Φ and Hamiltonian H) that satisfies the four pillars.
- By the meta-theorem, the existence of such an encoding guarantees that there is a finite deterministic algorithm (the D-RSP algorithm) that decides the conjecture.
- The proof of correctness of this algorithm for a given conjecture is therefore **reduced** to verifying the four pillars for that specific encoding. Once the pillars are verified, the meta-theorem provides the decidability and the certificate.

Thus, the Algorithmic Spectral Completeness Theorem acts as a **uniformity principle**: it ensures that the same D-RSP algorithm works for all problems that admit a spectral encoding, without needing to re-prove the convergence or extraction properties each time. In practice, for each of the thirty-three problems, we:

1. Construct the SAT instance Φ encoding the problem.
2. Build the spectral Hamiltonian $H = \hat{V} + \Delta$.
3. Verify the four pillars (P1–P4) using the generic theorems from the kernel (which are already proved for any hypercube graph and any diagonal non-negative potential).
4. Conclude via the meta-theorem that the D-RSP algorithm decides the conjecture, and that the extracted assignment (if satisfiable) provides a solution.

The following subsections detail how this scheme applies to each of the thirty-three resolved problems.

4.1 Problem 1: $P = NP$ (Series I)

– **CD** We encode a SAT instance into a spectral Hamiltonian on the hypercube. The four pillars are verified exactly as in Articles I.1–I.5. The D-RSP algorithm extracts a satisfying assignment in polynomial time. Hence SAT P, and by the Cook-Levin theorem $P = NP$.

4.2 Problem 2: Yang–Mills mass gap and confinement (Series II)

– **CD/FV** Lattice discretisation with Wilson action, spectral Hamiltonian, uniform spectral gap (Kithima Bridge). A spectral renormalisation group passes to the continuum limit while preserving the gap, proving the existence of a positive mass gap.

4.3 Problem 3: Beal (Series III)

– **EB** Using linear forms in logarithms (Stewart–Yu, Matveev), we obtain an explicit bound N_0 on any counterexample. Encoding the existence of a counterexample as a SAT instance of size polynomial in $\log N_0$ and running the spectral SAT solver proves unsatisfiability, hence no counterexample exists.

4.4 Problem 4: Riemann hypothesis (Series IV)

– **SRH** Construction of an explicit self-adjoint operator on a Hilbert space of prime numbers whose eigenvalues are exactly the imaginary parts of the non-trivial zeros of $\zeta(s)$. Self-adjointness forces all eigenvalues to be real, hence all zeros lie on the critical line. The Hilbert–Pólya conjecture is a corollary.

4.5 Problem 5: Goldbach (Series V)

– **CD** For each even n, encode the search for a Goldbach pair as a SAT instance. The four pillars are verified. The D-RSP algorithm extracts a prime pair.

4.6 Problem 6: Birch–Swinnerton-Dyer (BSD) (Series VI)

– **SRP** Construction of the adelic operator $M_E(s)$, isotypic compression to a finite matrix, encoding of the rank condition as a SAT instance, verification of the four pillars, D-RSP extracts the rank.

4.7 Problem 7: Singmaster (Series VII)

– **EB** Effective bound T_0, SAT encoding, D-RSP enumerates all solutions.

4.8 Problem 8: Dubner – twin prime conjecture (Series VIII)

– **FV** Asymptotic bound (circle method + Bombieri–Vinogradov), SAT encoding for even n up to N_0, D-RSP extracts a twin-prime pair.

4.9 Problem 9: Legendre (Series IX)

– **CD** Direct SAT encoding, D-RSP extracts a prime between n^2 and $(n+1)^2$.

4.10 Problem 10: Fermat–Catalan (Series X)

– **EB** Linear forms in logarithms bound exponents, SAT encoding, D-RSP proves unsatisfiability.

4.11 Problem 11: Lemoine (Series XI)

– **FV** Circle method asymptotic bound, SAT encoding for $n \leq N_0$, D-RSP extracts representation $n = p + 2q$.

4.12 Problem 12: Oppermann (Series XII)

– **CD** Similar to Legendre, D-RSP extracts two primes in intervals.

4.13 Problem 13: abc (Series XIII)

– **EB** Effective bound (Stewart–Yu), SAT encoding, D-RSP proves unsatisfiability; Hall conjecture as corollary.

4.14 Problem 14: Kithima-Landau – fourth Landau problem (Series XIV)

– **FV** Chen-type sieve bound, SAT encoding for $n \leq N_0$, D-RSP extracts prime of form $n^2 + 1$.

4.15 Problem 15: Hadamard matrices (Series XV)

– **CD** SAT encoding of orthogonality conditions for a ± 1 matrix, D-RSP extracts a Hadamard matrix.

4.16 Problem 16: Williamson matrices (Series XVI)

– **CD** Similar encoding for four symmetric ± 1 matrices, D-RSP extraction.

4.17 Problem 17: Maximal determinant (Series XVII)

– **CD** Binary search on determinant value, SAT encoding for each threshold, D-RSP decides.

4.18 Problem 18: Goormaghtigh (Series XVIII)

– **EB** Linear forms in logarithms bound variables, SAT encoding, D-RSP proves unsatisfiability.

4.19 Problem 19: Pollock tetrahedral (Series XIX)

– **FV** Circle method bound, SAT encoding for $N \leq N_0$, D-RSP extracts decomposition.

4.20 Problem 20: Pollock octahedral (Series XX)

– **FV** Parametric formulas bound, SAT encoding, D-RSP extracts decomposition.

4.21 Problem 21: Brocard (Series XXI)

– **FV** Numerical bound 10^9, SAT encoding for $n \leq 10^9$, D-RSP extracts known solutions.

4.22 Problem 22: 1/3–2/3 conjecture (Kislitsyn) (Series XXII)

– **CD** SAT encoding of balanced pair in poset, D-RSP extracts pair.

4.23 Problem 23: Pillai (Series XXIII)

– **EB** Effective bound (Bennett, Bugeaud–Mignotte–Siksek), SAT encoding, D-RSP proves unsatisfiability (or lists finite exceptions).

4.24 Problem 24: n-conjecture (Series XXIV)

– **EB** Generalisation of abc effective bound, SAT encoding, D-RSP proves unsatisfiability.

4.25 Problem 25: Vizing (Series XXV)

– **CD** Asymptotic theorem (Clark–Suen) bounds graph sizes, SAT encoding for small graphs, D-RSP proves unsatisfiability.

4.26 Problem 26: Erdős–Hajnal (Series XXVI)

– **EB** Explicit constants $\varepsilon(H)$ and $n_0(H)$, SAT encoding for $n < n_0(H)$, D-RSP proves unsatisfiability.

4.27 Problem 27: Gilbert–Pollak (Series XXVII)

– **FV** Discretisation + asymptotic bound, SAT encoding for $n \leq N_0$, D-RSP proves unsatisfiability.

4.28 Problem 28: Sumner (Series XXVIII)

– **FV** Kühn–Osthus asymptotic theorem, SAT encoding for $n < n_0$, D-RSP proves unsatisfiability.

4.29 Problem 29: Leopoldt (Series XXIX)

– **FD** Spectral transfer operator on p-adic étale cohomology, verification of FD pillars, functional D-RSP extracts eigenvalue; positivity proves the conjecture.

4.30 Problem 30: Kummer–Vandiver (Series XXX)

– **FD** Transfer operator on $K_4(\mathbb{Z})$ (modular symbols), self-adjointness, constant spectral gap (Quillen–Lichtenbaum), wavelet area law, functional D-RSP proves $p \nmid h_p^+$.

4.31 Problem 31: Fuglede (dimensions 1–4) (Series XXXI)

– **SUTL** Differential operator U_Ω on $L^2(\Omega)$, Kato–Osborn compression to finite matrix, SAT encoding of lattice and pure point conditions, D-RSP decides equivalence.

4.32 Problem 32: Barnette (Series XXXII)

– **SHS** Transfer operator T_G on spanning cycles, isotypic compression, SAT encoding of Hamiltonian cycle existence, D-RSP extracts cycle for all cubic bipartite planar graphs.

4.33 Problem 33: Infinitude of prime families (cousins, sexy, etc.) (Series XXXIII)

– **FV** For each even gap d, sieve result gives asymptotic bound $N_0(d)$, SAT encoding for even $n \leq N_0(d)$, D-RSP extracts prime pairs; hence infinitely many such primes.

5 Comparison with other spectral approaches

The theorem applies only to problems that admit an encoding satisfying the four pillars. Among advanced strategies, the following are complete and proven: SRH (Riemann), SRP (BSD), SUTL (Fuglede), SHS (Barnette). The Hilbert–Pólya conjecture is a corollary of SRH (Article IV.7). The strategies FM, DUST, SFF, Backward Transfer, CTP remain open. The Navier-Stokes spectral approach (using FD) is incomplete and not counted among the resolved problems.

6 Instantiation for the thirty-three problems

All thirty-three problems satisfy the conditions of the theorem. The definitive list is given in Article0.9 (table with 33 entries).

7 Conclusion

The Theorem of Algorithmic Spectral Completeness establishes a clear, testable criterion for decidability via spectral methods. It separates problems that are amenable to the core strategies from those that require more specialised (and often still open) approaches. The theorem is a meta-theorem whose proof is included and follows directly from the Spectral Reduction Lemma. It now serves as the logical capstone of the foundational series (0.0–0.12). Together with the Spectral Reduction Lemma, it provides both a powerful tool and a unifying meta-statement that guarantees the decidability of any problem that can be captured by the four pillars.

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